

Menu Driven Apps

Loops, menus and the switch statement

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Topics list

- A simple menu using switch.
- ShopV3.0 – adding a menu.

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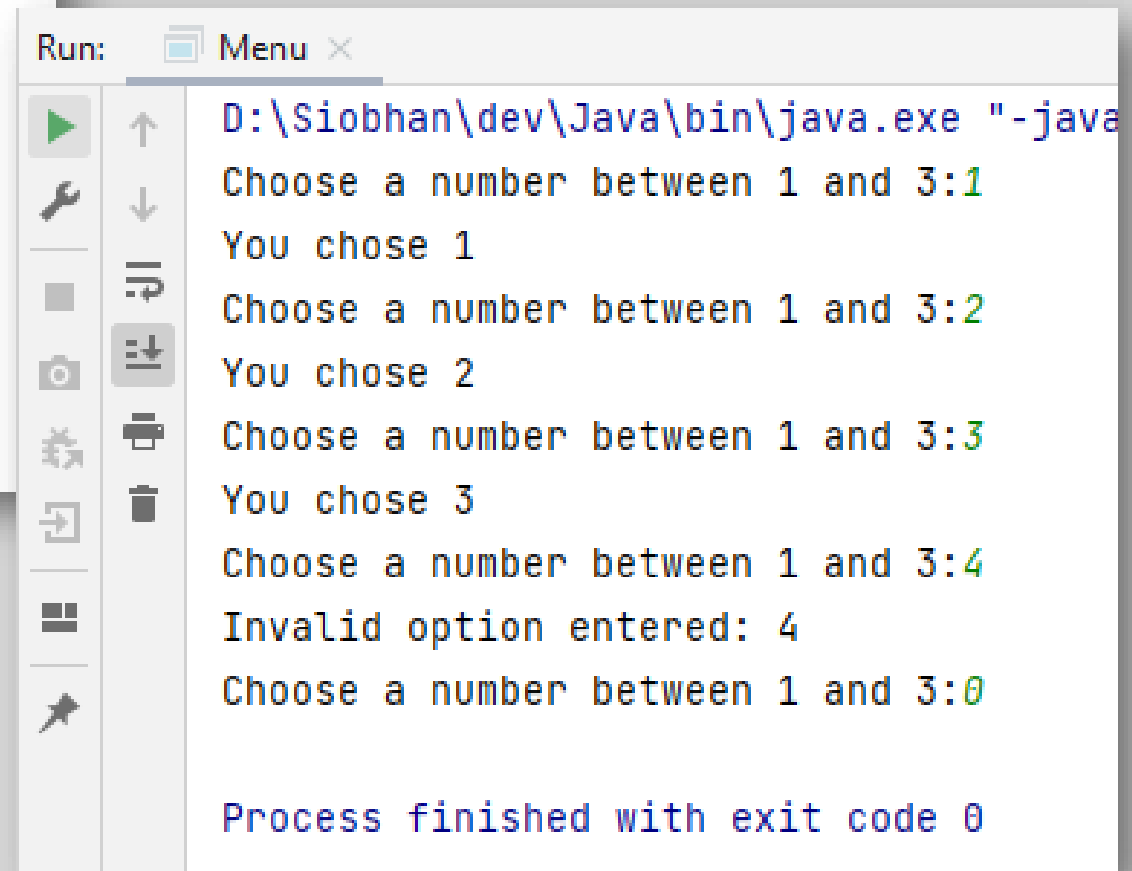
Now loop on the switch statement

Note the use of the Loop Control Variable, **option**.

```
private void runMenu(){  
    int option = ScannerInput.readNextInt( prompt: "Choose a number between 1 and 3:");  
    while (option != 0){  
        switch (option){  
            case 1 -> System.out.println("You chose 1");  
            case 2 -> System.out.println("You chose 2");  
            case 3 -> System.out.println("You chose 3");  
            default -> System.out.println("Invalid option entered: " + option);  
        }  
        option = ScannerInput.readNextInt( prompt: "Choose a number between 1 and 3:");  
    }  
}
```

This gives the following output

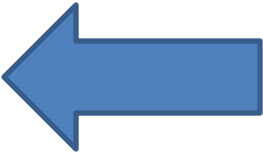
```
private void runMenu(){  
  
    int option = ScannerInput.readNextInt( prompt: "Choose a number between 1 and 3:");  
  
    while (option != 0){  
  
        switch (option){  
            case 1 -> System.out.println("You chose 1");  
            case 2 -> System.out.println("You chose 2");  
            case 3 -> System.out.println("You chose 3");  
            default -> System.out.println("Invalid option entered: " + option);  
        }  
  
        option = ScannerInput.readNextInt( prompt: "Choose a number between 1 and 3:");  
    }  
}
```



```
Run: Menu x  
D:\Siobhan\dev\Java\bin\java.exe "-java  
Choose a number between 1 and 3:1  
You chose 1  
Choose a number between 1 and 3:2  
You chose 2  
Choose a number between 1 and 3:3  
You chose 3  
Choose a number between 1 and 3:4  
Invalid option entered: 4  
Choose a number between 1 and 3:0  
  
Process finished with exit code 0
```

If you wanted more than one line of code associated with a case, just declare a code block:

```
private void runMenu(){  
  
    int option = ScannerInput.readNextInt(prompt: "Choose a number between 1 and 3:");  
  
    while (option != 0){  
  
        switch (option){  
            case 1 -> {  
                System.out.println("You chose 1");  
                System.out.println("You chose One");  
            }  
            case 2 -> System.out.println("You chose 2");  
            case 3 -> System.out.println("You chose 3");  
            default -> System.out.println("Invalid option entered: " + option);  
        }  
  
        option = ScannerInput.readNextInt(prompt: "Choose a number between 1 and 3:");  
    }  
}
```



Topics list

- A simple menu using switch.
- ShopV3.0 – adding a menu.

Adding a basic menu to Shop...

```
Shop Menu
```

```
-----
```

```
1) Add a product
```

```
2) List the Products
```

```
-----
```

```
3) List the current products
```

```
4) Display average product unit cost
```

```
5) Display cheapest product
```

```
6) List products that are more expensive than a given price
```

```
-----
```

```
0) Exit
```

```
==>>
```


Menu to be displayed...

Driver Class

```
private int mainMenu(){
    int option = ScannerInput.readNextInt( prompt: ""
        Shop Menu
        -----
        1) Add a product
        2) List the Products
        -----
        3) List the current products
        4) Display average product unit cost
        5) Display cheapest product
        6) List products that are more expensive than a given price
        -----
        0) Exit
        ==>>""");
    return option;
}
```

Shop Menu

```
-----
1) Add a product
2) List the Products
-----
3) List the current products
4) Display average product unit cost
5) Display cheapest product
6) List products that are more expensive than a given price
-----
0) Exit
==>>
```

Menu to be displayed...

```
private int mainMenu(){
    int option = ScannerInput.readNextInt( prompt: """
        Shop Menu
        -----
        1) Add a product
        2) List the Products
        -----
        3) List the current products
        4) Display average product unit cost
        5) Display cheapest product
        6) List products that are more expensive than a given price
        -----
        0) Exit
    """);
    return option;
}
```

Driver Class

We are using **Raw Strings** (the triple double quotes) to format the output for the menu.

Handling the menu input...

Driver Class

```
private void runMenu(){
    int option = mainMenu();

    while (option != 0){

        switch (option){
            case 1 -> addProduct();
            case 2 -> printProducts();
            case 3 -> printCurrentProducts();
            case 4 -> printAverageProductPrice();
            case 5 -> printCheapestProduct();
            case 6 -> printProductsAboveAPrice();
            default -> System.out.println("Invalid option entered: " + option);
        }

        //pause the program so that the user can read what we just printed to the terminal window
        ScannerInput.nextLine(prompt: "\nPress enter key to continue...");

        //display the main menu again
        option = mainMenu();
    }
}
```

Calling the menu on startup...

Driver Class

```
public class Driver{  
  
    private Store store = new Store();  
  
    public static void main(String[] args) {  
        new Driver();  
    }  
  
    public Driver() {  
        runMenu();  
    }  
}
```

Shop Menu

- ```

1) Add a product
2) List the Products

3) List the current products
4) Display average product unit cost
5) Display cheapest product
6) List products that are more expensive than a given price

0) Exit
```

==>>

# NOTE: A simple menu using the **older** format of the switch statement

---

```
public void run()
{
 System.out.println("Choose a number between 1 and 3");
 int choice = input.nextInt();

 switch(choice)
 {
 case 1:
 System.out.println("You chose 1");
 break;
 case 2:
 System.out.println("You chose 2");
 break;
 case 3:
 System.out.println("You chose 3");
 break;
 default:
 System.out.println("You chose an invalid number");
 break;
 }
}
```

A **break** statement is needed after each case, otherwise execution “falls” into the next statement regardless.

# Questions?

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